

6. Project Testing Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Verifying the generation workflow, routes, interface, and export

1. Testing Objective

Testing checks that the integrated ComicCraft application can accept user input, generate comic content, display the result, and export a PDF. The supplied guide describes local testing and verification; it does not provide a formal test report or measured results.

2. Environment and Server Check

Confirm that the virtual environment is active and required packages are installed. Start the application using ``uvicorn app.main:app --reload`` from the project root and confirm that the local server starts.

3. Homepage and Input Checks

Open 127.0.0.1:8000 and verify that the homepage loads. Check that the story prompt, character name, setting, tone, and art-style fields are visible and can be submitted.

4. Form-Based Generation Check

Submit a sample story through `/generate`. Verify the route receives the form fields and invokes the outline, story, image, layout, and PDF functions in the documented sequence.

5. JSON API Check

Open `/docs` and test `/generate-comic/json` with a JSON request matching the `PromptRequest` schema. Verify that the endpoint returns the documented layout data and PDF path.

6. Image Generation Check

Use `/test-image` to test image generation with a direct prompt. Confirm that the function produces an image and saves it in the `static/panels` directory.

7. Preview Check

Review `comic_preview.html` and confirm that panels appear sequentially with their titles, illustrations, scene descriptions, captions, and narration. Check that each image corresponds to its panel.

8. PDF Export Check

Use the download action on the preview page. Verify that a PDF is generated and that panel images and narrative text are included on the pages. Confirm that the application displays the export-success page.

9. Error and Configuration Checks

Check that missing or invalid request data is handled through validation and that unexpected failures are handled by the documented exception mechanisms. Confirm that API credentials are supplied through environment variables and are not exposed in page content.

10. Testing Outcome

The phase should record pass/fail observations for each check, along with any defects and retest results. The guide describes the testing workflow but does not state that every test above has been executed or passed.